

AP Statistics MCQ Practice Exam 09

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AP Statistics MCQ Practice Exam 9

PART A, QUESTIONS

APSTAT-MCQ9-001

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A dataset of 10 observations has mean 20. One observation, 29, is removed. What is the mean of the remaining 9 observations?

- A) 19.0
- B) 18.0
- C) 18.9
- D) 17.1

APSTAT-MCQ9-002

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A city survey estimates that between 62% and 70% of all households use a municipal service, at 95% confidence. Which public report is statistically appropriate?

- A) The true percentage changes from 62% to 70% in 95% of years.
- B) Exactly 95% of sampled households have usage rates between 62% and 70%.
- C) Based on the sampling method, 62% to 70% is a plausible range for the percentage of all city households using the service.
- D) There is a 95% chance that every nonsampled household uses the service.

APSTAT-MCQ9-003

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

A fair eight-sided die is labeled 1 through 8. Given that the result is greater than 3, what is the probability that it is even?

- A) $\frac{3}{5}$
- B) $\frac{1}{2}$
- C) $\frac{3}{8}$
- D) $\frac{5}{8}$

APSTAT-MCQ9-004

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Collect Data

Prompt:

Two laboratories will compare mean readings from two measurement methods. Which design most effectively removes specimen-to-specimen variation from the method comparison?

- A) Use different convenient specimens in the two laboratories.
- B) Let each laboratory choose specimens expected to favor its method.
- C) Apply both methods to each randomly selected specimen, randomizing method order when order could matter.
- D) Compare one measurement from each method without recording the specimen.

APSTAT-MCQ9-005

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Formulate Questions

Prompt:

A marine biologist wants to use water temperature to predict the number of feeding visits at individual tide-pool observation sessions. Which question most clearly defines the intended regression analysis and its observational unit?

- A) What is the mean water temperature across all sessions, ignoring feeding visits?
- B) Do warm sessions and cool sessions receive the same categorical label?
- C) How many feeding visits occurred at all tide pools combined?
- D) How does the number of feeding visits per session tend to change with water temperature measured during that same session?

APSTAT-MCQ9-006

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A survey contains two policy questions, and researchers are concerned that the first question may influence answers to the second. Which design best assesses a question-order effect?

- A) Ask interviewers which order they believe is less biased.
- B) Give every respondent order AB so administration is consistent.
- C) Let respondents choose the question they answer first.
- D) Randomly assign respondents to receive the questions in order AB or BA and compare responses.

APSTAT-MCQ9-007

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A 95% confidence interval for the population proportion p of users who enable a feature is $(0.62, 0.70)$. Let $q = 1 - p$ be the population proportion who do not enable it. What is the corresponding 95% confidence interval for q ?

- A) $(0.38, 0.30)$
- B) $(0.30, 0.38)$
- C) $(0.62, 0.70)$
- D) $(-0.70, -0.62)$

APSTAT-MCQ9-008

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

A random variable X has variance 0. Which statement must be true?

- A) X is normally distributed.
- B) X equals one constant value with probability 1.
- C) X has mean 0.

D) X takes positive and negative values with equal probability.

APSTAT-MCQ9-009

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A researcher tests $H_0: \mu_X = 40$ against $H_a: \mu_X > 40$ and obtains $t = 2.10$. The variable is redefined as $Y = 100 - X$. Which statement gives the equivalent test using Y?

- A) $H_0: \mu_Y = 40$ against $H_a: \mu_Y > 40$, with $t = 2.10$
- B) $H_0: \mu_Y = 60$ against $H_a: \mu_Y > 60$, with $t = -2.10$
- C) $H_0: \mu_Y = 60$ against $H_a: \mu_Y < 60$, with $t = -2.10$ and the same one-sided p-value
- D) $H_0: \mu_Y = 60$ against $H_a: \mu_Y < 60$, with $t = 2.10$ and a different p-value

APSTAT-MCQ9-010

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A histogram displays exact frequencies for intervals but not the individual observations within each interval. Which quantity can always be determined exactly from the histogram?

- A) The sample standard deviation
- B) The sample mean
- C) The sample median
- D) The total number of observations

APSTAT-MCQ9-011

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

Four residuals from a fitted regression line are -2 , 1 , 0 , and 3 . What is the sum of squared residuals for these four observations?

- A) 6
- B) 14

C) 10

D) 36

APSTAT-MCQ9-012

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Formulate Questions

Prompt:

A researcher wants to estimate how the population proportion using automatic payment differs between customers under age 30 and customers age 30 or older. Which parameter matches the goal?

A) $p_{\text{under30}} - p_{\text{30plus}}$

B) $\hat{p}_{\text{under30}} - \hat{p}_{\text{30plus}}$

C) $\mu_{\text{under30}} - \mu_{\text{30plus}}$

D) The number of sampled customers using automatic payment

APSTAT-MCQ9-013

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Collect Data

Prompt:

A theater row has 20 labeled seats, and four friends will receive four distinct seats at random. Which simulation plan best estimates the probability that no two friends receive adjacent seats?

A) Generate four integers from 1 to 20 with replacement and count trials with no consecutive values.

B) Choose one set of four nonadjacent seats and repeat that same assignment many times.

C) Generate the number of adjacent seat pairs directly from a normal distribution.

D) Randomly select four distinct seat numbers without replacement, record whether none are consecutive, and repeat for many trials.

APSTAT-MCQ9-014

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Interpret Results

Prompt:

The modal response category in a survey contains 35% of responses. Which statement is correct?

A) The category contains a majority because it is the mode.

- B) The category must also contain the median response.
- C) It is the most common category, but it does not contain a majority of responses.
- D) The remaining categories must each contain exactly 65% of responses.

APSTAT-MCQ9-015

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

For a t distribution with $df = 12$, the upper-tail areas beyond $t = 2.179$ and $t = 3.055$ are 0.025 and 0.005, respectively. A two-sided test produces $|t| = 2.50$. Which range contains the p -value?

- A) Greater than 0.10
- B) Between 0.005 and 0.025
- C) Between 0.05 and 0.10
- D) Between 0.01 and 0.05

APSTAT-MCQ9-016

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A one-proportion test of $H_0: p = 0.40$ against $H_a: p > 0.40$ has $\alpha = 0.05$ and power 0.80 when $p = 0.50$. Which pair of probabilities is correct?

- A) $P(\text{reject } H_0 \mid p = 0.40) = 0.95$ and $P(\text{fail to reject } H_0 \mid p = 0.50) = 0.80$
- B) $P(\text{reject } H_0 \mid p = 0.40) = 0.20$ and $P(\text{fail to reject } H_0 \mid p = 0.50) = 0.05$
- C) $P(\text{reject } H_0 \mid p = 0.40) = 0.05$ and $P(\text{fail to reject } H_0 \mid p = 0.50) = 0.20$
- D) $P(\text{reject } H_0 \mid p = 0.40) = 0.80$ and $P(\text{fail to reject } H_0 \mid p = 0.50) = 0.95$

APSTAT-MCQ9-017

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Formulate Questions

Prompt:

A skating association wants to describe variation in lap times for all competitors at a national event. Which parameter most directly matches that goal?

- A) The sample mean lap time among competitors timed by one volunteer
- B) The population median rank of the competitors
- C) The population standard deviation of competitors' lap times
- D) The proportion of sampled competitors with lap times below their sample mean

Questions APSTAT-MCQ9-018 to APSTAT-MCQ9-020 refer to the following stimulus.

The fill error X , in grams, for a packaging machine is normally distributed with mean 0.4 gram and standard deviation 1.2 grams. Errors on independently selected packages are independent. Use $\Phi(1.00) = 0.8413$, $\Phi(3.00) = 0.99865$, and the 10th-percentile standard normal value $z = -1.282$.

APSTAT-MCQ9-018

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For X normally distributed with mean 0.4 gram and standard deviation 1.2 grams, what is $P(X > 1.6 \text{ grams})$ for one randomly selected package?

- A) 0.0228
- B) 0.3413
- C) 0.8413
- D) 0.1587

APSTAT-MCQ9-019

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For a normal fill-error distribution with mean 0.4 gram and standard deviation 1.2 grams, what is the 10th percentile, approximately? Use $z = -1.282$.

- A) -1.14 grams
- B) -0.88 gram
- C) 0.12 gram
- D) 1.94 grams

APSTAT-MCQ9-020

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Interpret Results

Prompt:

For a random sample of 9 packages, $\bar{x}$ is normally distributed with mean 0.4 gram and standard deviation $1.2/\sqrt{9} = 0.4$ gram. What is $P(\bar{x} > 1.6 \text{ grams})$, and what does it describe?

- A) 0.1587, the chance that one package exceeds 1.6 grams
- B) 0.00135, the chance that the average error of 9 packages exceeds 1.6 grams
- C) 0.0228, the chance that all 9 packages exceed 1.6 grams
- D) 0.99865, the chance that the average error is below its mean

APSTAT-MCQ9-021

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

Changing measurement units makes a fitted line's positive slope numerically large, while its correlation remains $r = 0.20$. What should be concluded?

- A) The linear association became strong when the slope's numerical value increased.
- B) The relationship became negative because r is below 1.
- C) The predicted response rises with x , but the standardized linear association remains weak.
- D) A large positive slope and $r = 0.20$ are mathematically incompatible.

APSTAT-MCQ9-022

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Collect Data

Prompt:

A website compares response proportions for two invitation messages. Which design best supports a causal comparison?

- A) Randomly assign eligible users to receive message A or B, then compare response proportions.
- B) Let users select which message they prefer to receive.
- C) Send A to new users and B to returning users.
- D) Send message A in the morning and message B in the evening.

APSTAT-MCQ9-023

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

The dataset 2, 4, 6 has sample standard deviation 2. Each observation is then duplicated once, producing 2, 2, 4, 4, 6, 6. What is the sample standard deviation of the duplicated dataset, approximately?

- A) 2.00
- B) 1.79
- C) 2.53
- D) 4.00

APSTAT-MCQ9-024

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Formulate Questions

Prompt:

A random sample of households is measured before and after attic insulation is installed. Researchers want to estimate the population mean decrease in monthly energy use. Which procedure is appropriate?

- A) A one-sample t-interval for the within-household before-minus-after differences
- B) A two-sample t-interval treating before and after measurements as independent
- C) A one-proportion z-interval for households whose use decreases
- D) A one-sample t-interval using only the after-insulation measurements

APSTAT-MCQ9-025

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For a discrete random variable X , $E(X) = 3$ and $E(X^2) = 11$. Using $\text{Var}(X) = E(X^2) - [E(X)]^2$, what is the standard deviation of X ?

- A) 2
- B) $\sqrt{2}$
- C) 8

D) 11

APSTAT-MCQ9-026

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A quality team tests $H_0: p = 0.02$ after every additional 10 inspected items and stops the first time the p-value is below 0.05. It reports the final test as having $\alpha = 0.05$. What is the main problem with that claim?

- A) A proportion can be tested only after every item in the population is inspected.
- B) The p-value must equal the observed sample proportion.
- C) Stopping early always increases the estimated defect proportion to 1.
- D) Repeatedly looking and stopping after a small p-value can make the overall Type I error probability exceed 0.05.

APSTAT-MCQ9-027

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A software study compares two navigation layouts, and prior experience suggests that operating system strongly affects task time. Which design best controls this source of variation?

- A) Block participants by operating system, then randomly assign the two layouts within each block.
- B) Give one layout to all users of one operating system and the other layout to all remaining users.
- C) Allow each participant to choose the layout that appears easier.
- D) Ignore operating system and assign layouts in order of arrival without randomization.

APSTAT-MCQ9-028

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

An analyst defines paired differences as $d = \text{after} - \text{before}$ and obtains $t = -2.40$ with a two-sided p-value of 0.024 for $H_0: \mu_d = 0$. If the differences are instead defined as $d' = \text{before} - \text{after}$, what result

is obtained?

- A) $t = -2.40$ and $p\text{-value} = 0.976$
- B) $t = 2.40$ and $p\text{-value} = 0.048$
- C) $t = 2.40$ and $p\text{-value} = 0.024$
- D) $t = 0$ and $p\text{-value} = 1$

APSTAT-MCQ9-029

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Interpret Results

Prompt:

A study samples 10 students from each of 6 randomly selected class sections and compares a quantitative outcome between two programs. The analysis treats all 60 students as independent. Which assessment is most appropriate?

- A) The analysis is valid because 60 is greater than 30.
- B) The analysis overstates uncertainty because students in a section must have negative correlation.
- C) The analysis may understate the standard error because outcomes of students in the same section can be positively associated.
- D) The analysis becomes a paired t procedure because each section contributes 10 students.

APSTAT-MCQ9-030

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A value of 72 has z -score 1.50 in a distribution with mean 60. What is the distribution's standard deviation?

- A) 6
- B) 12
- C) 18
- D) 8

APSTAT-MCQ9-031

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Formulate Questions

Prompt:

Software repeatedly generates batches of 10 observations from a specified population model. To approximate the sampling distribution of $\bar{x}$ for $n = 10$, which output from each batch should be saved?

- A) Its maximum observation
- B) Its arithmetic average
- C) The sign of its first observation
- D) The fixed population mean

APSTAT-MCQ9-032

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

A survey uses a probability sample, but some respondents systematically underreport a behavior. Which statement about a conventional confidence interval for the behavior's proportion is correct?

- A) The margin of error describes random sampling variability but does not account for systematic measurement bias.
- B) The confidence level automatically corrects inaccurate answers.
- C) Increasing confidence eliminates the underreporting.
- D) The interval is unbiased because the original sample was random.

APSTAT-MCQ9-033

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Analyze Data

Prompt:

A new observation is added to a dataset, and the new value lies strictly between the old minimum and maximum. Which measure must remain unchanged?

- A) The mean
- B) The median
- C) The standard deviation
- D) The range

APSTAT-MCQ9-034

Section: MCQ

Type: mcq

Unit: Unit 4, Inference for Quantitative Data: Means

Skill: Analyze Data

Prompt:

A sample of 16 observations consists entirely of the value 54. What happens when the usual one-sample t statistic is computed for testing $H_0: \mu = 50$?

- A) $t = 0$ because there is no sample variation.
- B) $t = 1$ because the sample has only one distinct value.
- C) $t = 16$ because every observation exceeds 50.
- D) No finite t statistic is produced because $s = 0$ makes the estimated standard error 0.

Questions APSTAT-MCQ9-035 to APSTAT-MCQ9-037 refer to the following stimulus.

A least-squares regression predicts weekly output y from training hours x . The slope is positive. For the response values, the total sum of squares is $SST = 250$, and the regression sum of squared residuals is $SSE = 90$.

APSTAT-MCQ9-035

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For a regression with $SST = 250$ and $SSE = 90$, what is r^2 ?

- A) 0.36
- B) 0.64
- C) 0.60
- D) 0.80

APSTAT-MCQ9-036

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Analyze Data

Prompt:

For the regression with $SST = 250$, $SSE = 90$, and $r^2 = 0.64$, suppose 10 units are added to every weekly output y while the training-hour values remain unchanged. What happens to SST , SSE , and r^2 ?

- A) SST increases, SSE stays fixed, and r^2 increases.
- B) SST and SSE remain unchanged, so r^2 remains 0.64.

C) SST and SSE each increase by 10, so r^2 becomes 0.80.

D) SST stays fixed, SSE decreases, and r^2 becomes 1.

APSTAT-MCQ9-037

Section: MCQ

Type: mcq

Unit: Unit 5, Regression Analysis

Skill: Interpret Results

Prompt:

For a regression with $SST = 250$ and $SSE = 90$, which interpretation is correct?

A) The line predicts every weekly output within 36% of its observed value.

B) Training hours cause exactly 64% of each worker's output.

C) The line accounts for 64% of the variation in weekly output, while 36% remains in squared residual variation.

D) The correlation equals $SSE/SST = 0.36$.

APSTAT-MCQ9-038

Section: MCQ

Type: mcq

Unit: Unit 2, Probability, Random Variables, and Probability Distributions

Skill: Analyze Data

Prompt:

For a Bernoulli random variable with success probability p , the variance is $p(1 - p)$. At which value of p is this variance largest?

A) 0.50

B) 0.25

C) 0.75

D) 1.00

APSTAT-MCQ9-039

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Analyze Data

Prompt:

A one-proportion test produces $z = 2.05$ for $H_a: p > p_0$. The standard normal upper-tail probability beyond 2.05 is 0.020. What is the p-value?

A) 0.020

- B) 0.500
- C) 0.040
- D) 0.980

APSTAT-MCQ9-040

Section: MCQ

Type: mcq

Unit: Unit 3, Inference for Categorical Data: Proportions

Skill: Interpret Results

Prompt:

Researchers conduct 20 unrelated tests whose null hypotheses are all true, each at $\alpha = 0.05$, and publicize only the smallest p-value. Which concern is most appropriate?

- A) Every test must have a p-value greater than 0.05 because all null hypotheses are true.
- B) Searching across many tests makes at least one false positive more likely than the 0.05 rate for a single prespecified test.
- C) The smallest p-value is the probability that all 20 null hypotheses are true.
- D) Publishing one result changes the sample proportion in every other test.

APSTAT-MCQ9-041

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A telephone survey uses multiple interviewers. Which step best reduces interviewer effects?

- A) Allow each interviewer to paraphrase questions freely.
- B) Assign each interviewer to only one demographic group.
- C) Train interviewers to use the same neutral script and monitor adherence to the protocol.
- D) Tell interviewers the responses the sponsor expects.

APSTAT-MCQ9-042

Section: MCQ

Type: mcq

Unit: Unit 1, Exploring One-Variable Data and Collecting Data

Skill: Collect Data

Prompt:

A beverage company wants to compare taste ratings for two labels placed on the same drink. Which design best isolates the label effect?

- A) Randomly assign coded labels to otherwise identical drink samples and blind tasters to the study hypothesis.
- B) Put label A on one flavor and label B on a different flavor.
- C) Let tasters choose which label they receive.
- D) Compare current customers shown label A with last year's customers shown label B.

PART B, ANSWER KEY AND EXPLANATIONS

APSTAT-MCQ9-001, Correct: A, Correct Answer: 19.0, Explanation: The original total is $10(20) = 200$. After removing 29, the remaining total is 171, so the new mean is $171/9 = 19.0$.

APSTAT-MCQ9-002, Correct: C, Correct Answer: Based on the sampling method, 62% to 70% is a plausible range for the percentage of all city households using the service., Explanation: The interval estimates one fixed citywide proportion using a method with a stated repeated-sampling confidence rate. It does not describe a time-varying parameter or the certainty of individual household behavior.

APSTAT-MCQ9-003, Correct: A, Correct Answer: $3/5$, Explanation: Given a result greater than 3, the possible outcomes are 4, 5, 6, 7, and 8. Three of those five outcomes—4, 6, and 8—are even, so the conditional probability is $3/5$.

APSTAT-MCQ9-004, Correct: C, Correct Answer: Apply both methods to each randomly selected specimen, randomizing method order when order could matter., Explanation: Using both methods on each specimen creates paired data and removes between-specimen variation from the method comparison. Randomizing order limits order effects.

APSTAT-MCQ9-005, Correct: D, Correct Answer: How does the number of feeding visits per session tend to change with water temperature measured during that same session?, Explanation: The question identifies two quantitative variables measured on the same observational unit, a tide-pool session, and makes temperature the explanatory variable used to predict the session's feeding-visit count.

APSTAT-MCQ9-006, Correct: D, Correct Answer: Randomly assign respondents to receive the questions in order AB or BA and compare responses., Explanation: Random assignment makes the order groups comparable, so systematic response differences can be attributed to question order for the respondents studied.

APSTAT-MCQ9-007, Correct: B, Correct Answer: (0.30, 0.38), Explanation: The transformation $q = 1 - p$ reverses the interval endpoints. The lower endpoint is $1 - 0.70 = 0.30$ and the upper endpoint is $1 - 0.62 = 0.38$; the confidence level remains 95%.

APSTAT-MCQ9-008, Correct: B, Correct Answer: X equals one constant value with probability 1., Explanation: Variance is the expected squared distance from the mean. It can be zero only when that distance is zero with probability 1, so X is constant at its mean.

APSTAT-MCQ9-009, Correct: C, Correct Answer: $H_0: \mu_Y = 60$ against $H_a: \mu_Y < 60$, with $t = -2.10$ and the same one-sided p-value, Explanation: The transformation reverses direction: $\mu_X = 40$ corresponds to $\mu_Y = 60$, and $\mu_X > 40$ corresponds to $\mu_Y < 60$. Multiplication by -1 reverses the sign of t , while the equivalent lower-tail area equals the original upper-tail p-value.

APSTAT-MCQ9-010, Correct: D, Correct Answer: The total number of observations, Explanation: Exact bar frequencies can be summed to obtain n . Without the individual positions inside bins, the exact mean, median, and standard deviation generally cannot be recovered.

APSTAT-MCQ9-011, Correct: B, Correct Answer: 14, Explanation: SSE is the sum of squared residuals: $(-2)^2 + 1^2 + 0^2 + 3^2 = 4 + 1 + 0 + 9 = 14$.

APSTAT-MCQ9-012, Correct: A, Correct Answer: $p_{\text{under30}} - p_{\text{30plus}}$, Explanation: The target is a difference between two population proportions. The corresponding sample proportions are statistics used to estimate this parameter.

APSTAT-MCQ9-013, Correct: D, Correct Answer: Randomly select four distinct seat numbers without replacement, record whether none are consecutive, and repeat for many trials., Explanation: Each trial must reproduce four distinct random seat assignments. Recording the no-consecutive-seats indicator and repeating many times makes its relative frequency an estimate of the desired probability.

APSTAT-MCQ9-014, Correct: C, Correct Answer: It is the most common category, but it does not contain a majority of responses., Explanation: A mode is the category with the greatest frequency. A majority requires more than 50%, and 35% is not a majority.

APSTAT-MCQ9-015, Correct: D, Correct Answer: Between 0.01 and 0.05, Explanation: Because $2.179 < 2.50 < 3.055$, one upper-tail area is between 0.005 and 0.025. A two-sided p -value doubles that area, placing it between 0.01 and 0.05.

APSTAT-MCQ9-016, Correct: C, Correct Answer: $P(\text{reject } H_0 \mid p = 0.40) = 0.05$ and $P(\text{fail to reject } H_0 \mid p = 0.50) = 0.20$, Explanation: The significance level is the rejection probability when the null value is true, so the first probability is 0.05. Power is 0.80 at $p = 0.50$, so the Type II probability there is $1 - 0.80 = 0.20$.

APSTAT-MCQ9-017, Correct: C, Correct Answer: The population standard deviation of competitors' lap times, Explanation: The target is variability in one quantitative variable for the full population of competitors. The population standard deviation, σ , directly describes that spread; a sample mean and a proportion target different features.

APSTAT-MCQ9-018, Correct: D, Correct Answer: 0.1587, Explanation: Standardizing gives $z = (1.6 - 0.4)/1.2 = 1.00$. Therefore $P(X > 1.6) = 1 - \Phi(1.00) = 1 - 0.8413 = 0.1587$.

APSTAT-MCQ9-019, Correct: A, Correct Answer: -1.14 grams, Explanation: Convert the supplied 10th-percentile z -value to the original scale: $x = 0.4 + (-1.282)(1.2) = -1.1384$ grams, approximately -1.14 grams.

APSTAT-MCQ9-020, Correct: B, Correct Answer: 0.00135, the chance that the average error of 9 packages exceeds 1.6 grams, Explanation: The standardized sample mean is $z = (1.6 - 0.4)/0.4 = 3.00$. Thus $P(\bar{x} > 1.6) = 1 - 0.99865 = 0.00135$, a probability about the nine-package average rather than one package.

APSTAT-MCQ9-021, Correct: C, Correct Answer: The predicted response rises with x , but the standardized linear association remains weak., Explanation: The positive slope gives the direction of prediction, while $r = 0.20$ indicates weak linear association. Changing units can change the numerical

slope but not the correlation.

APSTAT-MCQ9-022, Correct: A, Correct Answer: Randomly assign eligible users to receive message A or B, then compare response proportions., Explanation: Random assignment balances other user characteristics across messages and supports a causal comparison of response proportions for eligible users in the experiment.

APSTAT-MCQ9-023, Correct: B, Correct Answer: 1.79, Explanation: The duplicated values still have mean 4, and their squared deviations sum to 16. The sample variance is $16/(6 - 1) = 3.2$, so the sample standard deviation is $\sqrt{3.2} \approx 1.79$; duplicating values changes the $n - 1$ divisor.

APSTAT-MCQ9-024, Correct: A, Correct Answer: A one-sample t-interval for the within-household before-minus-after differences, Explanation: Before and after measurements come from the same household, so one quantitative difference is calculated per household and analyzed with a one-sample t procedure.

APSTAT-MCQ9-025, Correct: B, Correct Answer: $\sqrt{2}$, Explanation: $\text{Var}(X) = 11 - 3^2 = 2$. Standard deviation is the nonnegative square root of variance, so $\sigma_X = \sqrt{2}$.

APSTAT-MCQ9-026, Correct: D, Correct Answer: Repeatedly looking and stopping after a small p-value can make the overall Type I error probability exceed 0.05., Explanation: The 0.05 level applies to one prespecified analysis under its stated sampling plan. Repeated opportunities to stop on a rare small p-value increase the chance of at least one false rejection unless the procedure adjusts for sequential monitoring.

APSTAT-MCQ9-027, Correct: A, Correct Answer: Block participants by operating system, then randomly assign the two layouts within each block., Explanation: Blocking compares layouts among participants sharing the important operating-system characteristic. Random assignment within blocks prevents layout from being confounded with operating system and supports a causal comparison.

APSTAT-MCQ9-028, Correct: C, Correct Answer: $t = 2.40$ and p-value = 0.024, Explanation: Reversing every paired difference multiplies the sample mean difference and t statistic by -1 . It does not change $|t|$, so a two-sided p-value remains 0.024.

APSTAT-MCQ9-029, Correct: C, Correct Answer: The analysis may understate the standard error because outcomes of students in the same section can be positively associated., Explanation: Students sampled from the same class section form clusters and need not provide independent information. Ignoring positive within-section association treats the effective amount of information as too large and can make the standard error too small.

APSTAT-MCQ9-030, Correct: D, Correct Answer: 8, Explanation: Use $z = (x - \mu)/\sigma$. Substituting gives $1.50 = (72 - 60)/\sigma = 12/\sigma$, so $\sigma = 12/1.50 = 8$.

APSTAT-MCQ9-031, Correct: B, Correct Answer: Its arithmetic average, Explanation: Each generated batch represents one sample of size 10, so saving its average produces one realization of $\bar{x}$. Repeating this builds an approximation to the sampling distribution.

APSTAT-MCQ9-032, Correct: A, Correct Answer: The margin of error describes random sampling variability but does not account for systematic measurement bias., Explanation: Probability sampling

supports the sampling-error calculation, but consistently inaccurate self-reports can shift the estimate and interval away from the true population proportion.

APSTAT-MCQ9-033, Correct: D, Correct Answer: The range, Explanation: A value strictly inside the old minimum and maximum changes neither endpoint, so the range remains unchanged. The mean, median, and standard deviation can change.

APSTAT-MCQ9-034, Correct: D, Correct Answer: No finite t statistic is produced because $s = 0$ makes the estimated standard error 0., Explanation: With all observations equal, $s = 0$ and $s/\sqrt{n} = 0$. The numerator $54 - 50$ is nonzero, so the usual t ratio requires division by zero and is not a finite statistic; the standard t procedure is inappropriate.

APSTAT-MCQ9-035, Correct: B, Correct Answer: 0.64, Explanation: $r^2 = 1 - \text{SSE}/\text{SST} = 1 - 90/250 = 1 - 0.36 = 0.64$.

APSTAT-MCQ9-036, Correct: B, Correct Answer: SST and SSE remain unchanged, so r^2 remains 0.64., Explanation: Adding a constant to every response shifts both observed and fitted values equally. Deviations from $\bar{y}$ and residuals are unchanged, so SST, SSE, and their ratio—and therefore r^2 —remain unchanged.

APSTAT-MCQ9-037, Correct: C, Correct Answer: The line accounts for 64% of the variation in weekly output, while 36% remains in squared residual variation., Explanation: The unexplained fraction is $\text{SSE}/\text{SST} = 90/250 = 0.36$, so the explained fraction is $1 - 0.36 = 0.64$. This describes variation, not individual causal percentages.

APSTAT-MCQ9-038, Correct: A, Correct Answer: 0.50, Explanation: The product $p(1 - p)$ is symmetric around 0.50 and reaches its maximum value 0.25 at $p = 0.50$.

APSTAT-MCQ9-039, Correct: A, Correct Answer: 0.020, Explanation: Because H_a is right-tailed, the p -value is the upper-tail probability beyond the observed positive z statistic. The supplied area is 0.020 and is not doubled.

APSTAT-MCQ9-040, Correct: B, Correct Answer: Searching across many tests makes at least one false positive more likely than the 0.05 rate for a single prespecified test., Explanation: Each true-null test has a 0.05 false-positive probability, but selecting the smallest result from many tests creates multiple opportunities for a false rejection. The highlighted p -value does not account for that search.

APSTAT-MCQ9-041, Correct: C, Correct Answer: Train interviewers to use the same neutral script and monitor adherence to the protocol., Explanation: Standardized neutral wording and monitoring reduce systematic differences caused by interviewer delivery or expectations.

APSTAT-MCQ9-042, Correct: A, Correct Answer: Randomly assign coded labels to otherwise identical drink samples and blind tasters to the study hypothesis., Explanation: Keeping the drink identical isolates label as the treatment, random assignment creates comparable groups, and blinding reduces expectation effects.

SERIES DO-NOT-REPEAT BANK UPDATE, APSTAT-MCQ9

- APSTAT-MCQ9-001 to 005: Unit 1 removes value 29 from ten observations with mean 20 to recover a nine-observation mean of 19; Unit 3 interprets a 95% municipal-service proportion interval (0.62, 0.70); Unit 2 conditions an eight-sided die on results above 3; Unit 4 applies two measurement methods to each random specimen with randomized order; Unit 5 formulates a tide-pool session regression question pairing water temperature with the same session's feeding-visit count. Misconceptions include failing to adjust the total, assigning posterior probability to a fixed parameter, retaining the unconditional die denominator, ignoring specimen pairing, and using unpaired totals instead of observational-unit data.
- APSTAT-MCQ9-006 to 010: Unit 1 randomizes AB versus BA survey-question order; Unit 3 transforms a 95% interval (0.62, 0.70) for p into (0.30, 0.38) for $q = 1 - p$; Unit 2 variance zero implies a constant random variable; Unit 4 the reflection $Y = 100 - X$ converts a right-tailed $t = 2.10$ into an equivalent left-tailed $t = -2.10$; Unit 1 an exact-frequency histogram determines n but not exact within-bin summaries. Misconceptions include using one fixed order, failing to reverse complement-proportion endpoints, requiring zero mean rather than zero spread, failing to reverse the alternative and statistic under reflection, and treating bin midpoints as raw data.
- APSTAT-MCQ9-011 to 015: Unit 5 residuals $-2, 1, 0,$ and 3 give $SSE = 14$; Unit 3 an age-group automatic-payment comparison defines $p_{\text{under30}} - p_{\text{30plus}}$; Unit 2 a 20-seat theater simulation samples four distinct seat numbers without replacement and records adjacency; Unit 1 a 35% modal category is a plurality but not a majority; Unit 4 supplied $df = 12$ t -tail cutoffs bracket a two-sided p -value for $|t| = 2.50$ between 0.01 and 0.05. Misconceptions include summing residuals instead of squares, using statistics as parameters, simulating distinct seats with replacement, equating mode with majority, and failing to double the bracketed one-tail areas.
- APSTAT-MCQ9-016 to 020: Unit 3 $\alpha = 0.05$ and power 0.80 are translated into a null rejection probability 0.05 and alternative nonrejection probability 0.20; Unit 1 population variability in national-event lap times targets σ ; Unit 2 a normal packaging-error set with $\mu = 0.4$ and $\sigma = 1.2$ finds an upper-tail probability at $z = 1$, converts a 10th-percentile z -value to grams, and evaluates a nine-package sample mean at $z = 3$. Misconceptions include confusing power with Type II probability, selecting a sample mean instead of a population spread parameter, using the wrong normal tail, leaving a percentile in z units, and using σ rather than $\sigma/\sqrt{n}$ for $\bar{x}$.
- APSTAT-MCQ9-021 to 025: Unit 5 distinguishes positive regression slope from weak correlation $r = 0.20$; Unit 3 randomly assigns website invitation messages for a causal binary-response comparison; Unit 1 duplicating values 2, 4, and 6 once changes the sample standard deviation to $\sqrt{(16/5)} \approx 1.79$; Unit 4 household pre/post insulation data use a one-sample t interval on differences; Unit 2 $E(X) = 3$ and $E(X^2) = 11$ give $\text{Var}(X) = 2$ and $\sigma_X = \sqrt{2}$. Misconceptions include treating slope magnitude as association strength, confounding message with user type or time, assuming replicated values leave the $n - 1$ divisor effect unchanged, discarding household pairing, and reporting variance rather than its square root as standard deviation.
- APSTAT-MCQ9-026 to 030: Unit 3 repeated interim testing with stopping at $p < 0.05$ inflates the overall Type I error; Unit 1 a navigation-layout experiment blocks by operating system and randomizes within blocks; Unit 4 reversing paired-difference direction changes t from -2.40 to 2.40 but preserves a two-sided p -value; Unit 4 clustered students within six class sections warn that an independence analysis can understate SE; Unit 1 reverses $z = (72 - 60)/\sigma = 1.50$ to obtain $\sigma = 8$. Misconceptions include treating α as valid under unplanned optional stopping, confounding layout with operating system, changing a two-sided p -value when signs reverse, counting clustered students as independent, and multiplying rather than dividing when solving a z -score formula.
- APSTAT-MCQ9-031 to 035: Unit 2 repeated generated batches of size 10 save each arithmetic average to approximate the sampling distribution of $\bar{x}$; Unit 3 a proportion interval's margin omits systematic self-report bias; Unit 1 adding an interior observation preserves the range but not necessarily other summaries; Unit 4 sixteen identical observations at 54 give $s = 0$ and no finite t statistic against μ_0

= 50; Unit 5 an SST = 250 and SSE = 90 regression stimulus calculates $r^2 = 0.64$. Misconceptions include storing maxima or the fixed population mean, expecting confidence to repair measurement bias, assuming all summaries remain fixed, interpreting zero sample spread as $t = 0$ despite a nonzero numerator, and reporting SSE/SST as explained variation.

· APSTAT-MCQ9-036 to 040: Unit 5 the SST/SSE stimulus shows that adding 10 to every response preserves SST, SSE, and r^2 , then interprets 64% explained versus 36% residual variation; Unit 2 Bernoulli variance $p(1 - p)$ is maximized at $p = 0.50$; Unit 3 a right-tailed $z = 2.05$ uses upper-tail probability 0.020; Unit 3 selecting the smallest p-value from 20 unrelated true-null tests raises the chance of a false positive. Misconceptions include changing sums of squares under a location shift, treating r^2 as an individual causal share, maximizing Bernoulli spread near an endpoint, doubling a one-sided p-value, and treating a selected minimum p-value as if it came from one prespecified test.

· APSTAT-MCQ9-041 to 042: Unit 1 standardized neutral interviewer scripts reduce interviewer effects; Unit 1 randomly assigned coded labels on identical drinks with blinded tasters isolate a label effect. Misconceptions include free paraphrasing or sponsor cues, and confounding label with flavor, self-selection, or historical groups.

· Unit counts for APSTAT-MCQ9: Unit 1 = 11; Unit 2 = 9; Unit 3 = 9; Unit 4 = 7; Unit 5 = 6. Skill counts: Formulate Questions = 5; Collect Data = 7; Analyze Data = 21; Interpret Results = 9. Stimulus sets: normal packaging-machine fill errors and a nine-package sample mean, APSTAT-MCQ9-018 to 020; regression decomposition from SST and SSE, APSTAT-MCQ9-035 to 037.